Jason White | Roboticist

jrw5690@gmail.com

Employment

Foundation Industries *2026 - present*

Senior Controls and Software Engineer (C++ & Python)

- Led controls and software development for a 22-DOF tendon driven hand, scaling it from a single motor prototype to a modular control stack deployed on humanoid hardware *[hardware, ~500Hz]* https://www.youtube.com/watch?v=Nhr7ZnFZYkA&t=1s
- Increased hardware communication rate from 20 Hz over UART to 250 Hz over a two CANFD lines through hardware abstraction, controller optimization, & firmware development
- Developed an highly accurate open loop position and torque dependent state estimator with a 10 microsecond solve time
- Improved the Phantom whole body inverse kinematics (optimization fixes) and Inspire hand grasp behavior (admittance control)
- Streamlined robot and hand calibration routines, zeroing routines, and added safety controllers
- Design decisions on tactile & position sensing, PCB components\locations, and communication protocol

Ghost Robotics *2024 - 2026*

Software and Controls Engineer (C++ & Python)

- Developed a system-wide locomotion controller using Model Predictive Control (MPC) and Pinocchio for real-time dynamic control *[hardware, ~500Hz]*
- Improved blind locomotion robustness on rugged terrain, reducing falls/flips from **7/20 to 1/20 trials** without reliance on perception *[hardware, ~500Hz]*
- Developed point cloud clustering and 2D height map filtering to enable terrain-aware locomotion *[hardware, ~90Hz]*
- Refactored core components of the codebase, eliminating blind locomotion low-level crashes and improving system reliability *[hardware, ~500Hz]*
- Designed filtering and tracking pipelines for manipulator input signals, eliminating drift and improving control stability *[hardware, ~500Hz]*
- Led system identification and design of next-generation leg actuators and drivetrain, achieving **3× torque output** and **50% mass reduction** *[hardware]*

Optimal Robotics Lab *2018 - 2023*

Graduate Research Assistant (Matlab & Python)

- Obstacle avoidance algorithm: for quadrotor & bipedal avoidance of fast-moving obstacles *[hardware, ~1kHz]* https://www.youtube.com/watch?v=cKKLuRdrqcQ
- Bipedal balancing controller: for dynamic terrain tested on Cassie *[hardware, >2kHz]* https://www.youtube.com/watch?v=q10Wg61H4UQ
- Real-time reactive planning and control for monopods (with flexible contact sequencing) *[simulated, ~500Hz]* https://www.youtube.com/watch?v=EquyU9cm-SM
- Quadrupedal stepping controller: using optimal control techniques (preliminary results) *[simulated, ~500Hz]* www.linkedin.com/feed/update/urn:li:activity:7122585426162249730/

Army Research Laboratory *2021 - 2022*

Graduate Research Fellow (Matlab implemented on hardware)

- Monopod planner through viscous media experimentation and controller development *[hardware, ~0.5Hz]*
- Iterative modeling and design of robotic leg

Technical Skills

Languages: C++, Python, MATLAB
Simulations: MuJoCo, Simulink, Bullet
Control Methods: Operational Space Control, Whole-Body Control, Forward/Inverse Dynamics, Joint Space Control, PD/PID Control, Force Control
Planning: Model Predictive Control, Direct Collocation, Sampling-based methods (RRT*, A*)
Optimization: Linear/Quadratic/Nonlinear Programming, Newton-Raphson

Awards, Fellowships, and Honors

2022: NASA Big Idea Challenge Finalist (ET-QUAD) *NASA*
2022: Best Locomotion Paper, IEEE International Conference on Robotics and Automation *IEEE*
2021: Research Fellowship Recipient, DEVCOM Army Research Laboratory *National*

Peer Reviewed Publications

Jason White, David Jay, Tianze Wang, and Christian Hubicki, "Avoiding Dynamic Obstacles with Real-time Motion Planning using Quadratic Programming for Varied Locomotion Modes," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 13626-13633) IEEE.

Jason White, Max Austin, Jonathan Clark, Christian Hubicki, and Jason Pusey. "Online Gait Optimization for Running in Resistive Media," in *2022 7th International Conference on Robotics and Automation Engineering (ICRAE)* (pp. 282-286). IEEE.

Jason White, Dylan Swart, and Christian Hubicki, "Force-based Control of Bipedal Balancing on Dynamic Terrain with the" Tallahassee Cassie" Robotic Platform," in *2020 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 6618-6624). IEEE.

Tianze Wang, **Jason White**, and Christian Hubicki, "Real-time Dynamic Bipedal Avoidance," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* IEEE.

Adwait Mane, Dylan Swart, **Jason White**, and Christian Hubicki. "Trajectory Optimization Formulation with Smooth Analytical Derivatives for Track-leg and Wheel-leg Ground Robots." in *2022 International Conference on Robotics and Automation (ICRA)* (pp. 5762-5768). IEEE.

Max Austin, John Nicholson, **Jason White**, Sean Gart, Ashley Chase, Jason Pusey, Christian Hubicki, Jonathan Clark, "Optimizing Dynamic Legged Locomotion in Mixed, Resistive Media," in *2022 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)* (pp. 1482-1488). IEEE.

Manuscripts & Posters

Jason White and Christian Hubicki, "Through-contact Monopod Motion Generation via MPC: Convex Formulation and Locomotion Analysis." 2024.

Jason White and Christian Hubicki, "Real-time Adversarial Obstacle Avoidance." 2023.

Jason White and Christian Hubicki "Toward Reduced-Order Modeling Approaches for Rapidly Reactive Locomotion." 2021 Dynamic Walking

Education

Florida State University
Ph.D. Mechanical Engineering - Legged Robot Control

M.S. Mechanical Engineering - Robotics, Dynamics, & Controls
Advisor: Christian Hubicki Ph.D.

Virginia Polytechnic Institute and State University

B.S. Civil Engineering - Structural Focus